

Lecture Tail Sum Formula, 尾数和公式.

$$E[R] = \sum_{i=0}^{\infty} P(R > i) \quad [7.1]$$

对于非负整数的随机变量 $[7.1] \Leftrightarrow E[R] = \sum_{k=0}^{\infty} k \cdot P(R=k) \Leftrightarrow E[R] = \sum_{i=0}^{\infty} P(R \geq i)$

易于证明, 只需展开即得 $\Rightarrow$ 以展开, 再以 k 为组, 变认于双重求和的重排.

"mcs" — 2015/5/18 — 1:43 — page 764 — #772

两者等价
(尾数和下)

1. 定义: 插值求和, Chapter 18 Random Variables

e.g. 插进 k 个时的概率

是 $\sum k \cdot P(R=k)$

his own hat back is $1/n$. There are many different probability distributions of hat permutations with this property, so we don't know enough about the distribution of G to calculate its expectation directly using equation (18.2) or (18.3). But linearity of expectation lets us sidestep this issue.

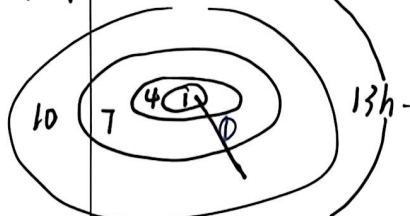
2. 尾数和: 尾和时程查

每时每刻时程查一次, 再累加

We'll use a standard, useful trick to apply linearity, namely, we'll express G as a sum of indicator variables. In particular, let G_i be an indicator for the event that the i th man gets his own hat. That is, $G_i = 1$ if the i th man gets his own hat, and $G_i = 0$ otherwise. The number of men that get their own hat is then the sum of these indicator random variables:

$$G = G_1 + G_2 + \dots + G_n. \quad (18.9)$$

来看: 设共 10 个可能



These indicator variables are *not* mutually independent. For example, if $n-1$ men all get their own hats, then the last man is certain to receive his own hat. But again, we don't need to worry about this dependence, since linearity holds regardless.

Since G_i is an indicator random variable, we know from Lemma 18.4.2 that

$$Ex[G_i] = Pr[G_i = 1] = 1/n. \quad (18.10)$$

By Linearity of Expectation and equation (18.9), this means that

$$\begin{aligned} Ex[G] &= Ex[G_1 + G_2 + \dots + G_n] \\ &= Ex[G_1] + Ex[G_2] + \dots + Ex[G_n] \\ &= \underbrace{\frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}}_n \\ &= 1. \end{aligned}$$

其中每一环均不平环, 即

4 & 7, 每个都为圆非环

现在我有 - 根线, 代表我等待

的时间: $0 = 1h + 4h + 7h + 10h$

这表时我有可能到达的时间

其概率为: $P_1 + P_4 + P_7 + P_{10}$

再分别乘上其大小, 复后得出

我真正可能的时间分布, 即求线.

概率由于均对其环归一化, 所以对其环归一化, 只对其自己有意义.

e.g. 对于 S_4 , 以知有 0.4 可能为 $4h$, 以有 0.6 可能为 $1h, 7h, 10h$ 等, 以自身为归一化.

以 0 为可能等待的区域, 那样求区间, 不同部分有不同“片数”, 即为权重 (待等待时间的可能性) $\Rightarrow$ 其概率即为附权

hat problem